

Guide to the Seller A Best-Response Plots

1 Model Objects Used in the Plots

The state in the simulation is the user's history with Seller A,

$$(S, F),$$

where S is the number of successes previously observed from Seller A and F is the number of failures. Because the user does not observe Seller A's hidden product choice, the user updates a misspecified single-product belief,

$$\theta \mid S, F \sim \text{Beta}(1 + S, 1 + F).$$

The probability that the Thompson-sampling user chooses Seller A is

$$\rho(S, F) = \Pr(\tilde{\theta} \geq p_0 \mid S, F), \quad \tilde{\theta} \sim \text{Beta}(1 + S, 1 + F),$$

where p_0 is the known success probability of Seller B.

For product $x \in \{1, 2\}$, define Seller A's product-specific action value as

$$Q_x(S, F) = R - c_x + \gamma [p_x V(S + 1, F) + (1 - p_x) V(S, F + 1)].$$

The optimal product at state (S, F) is the product with the larger Q value. Equivalently, product 2 is optimal when

$$Q_2(S, F) - Q_1(S, F) \geq 0.$$

Since

$$Q_2(S, F) - Q_1(S, F) = -(c_2 - c_1) + (p_2 - p_1) M_t(S, F),$$

where

$$M_t(S, F) = \gamma [V_{t+1}(S + 1, F) - V_{t+1}(S, F + 1)],$$

product 2 is optimal exactly when

$$M_t(S, F) \geq \frac{c_2 - c_1}{p_2 - p_1} = \frac{4}{3}.$$

If the undiscounted raw difference $D_t = V_{t+1}(S + 1, F) - V_{t+1}(S, F + 1)$ is reported instead, its threshold is $(c_2 - c_1)/[\gamma(p_2 - p_1)]$. The value-difference plots report M_t .

Exact discounted DP, gamma=0.980, T=700

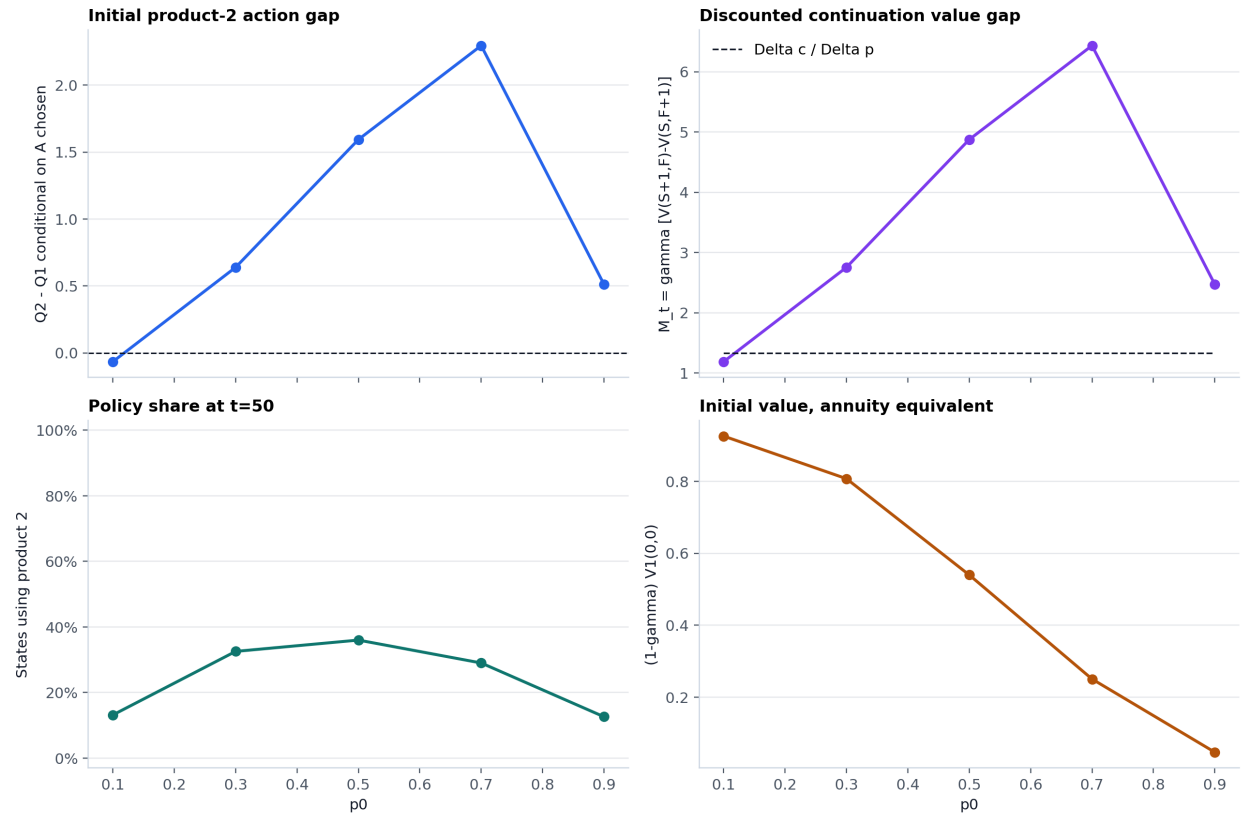


Figure 1: Summary of Seller A's best response as Seller B's known success probability p_0 varies.

2 Best Response as p_0 Varies

Figure 1 summarizes how Seller A's incentives change with p_0 .

- The top-left panel plots $Q_2(0,0) - Q_1(0,0)$ at the initial state. Positive values mean product 2 is optimal at the prior.
- The top-right panel reports $M_1(0,0)$ and the $4/3$ threshold.
- The bottom-left panel reports the share of reachable states using product 2 at the displayed policy period.
- The bottom-right panel reports the annuity-equivalent initial value $(1 - \gamma)V_1(0,0)$.

The key pattern is hump-shaped: product 2 is most attractive when Seller A is neither safely ahead nor hopelessly behind. If p_0 is very low, Seller B is weak, so Seller A can often save cost with product 1. If p_0 is very high, the chance of future demand from A is too low to justify much investment. For intermediate p_0 , a success can meaningfully improve future demand, so product 2 is more valuable.

3 Policy Heatmaps in Count Space

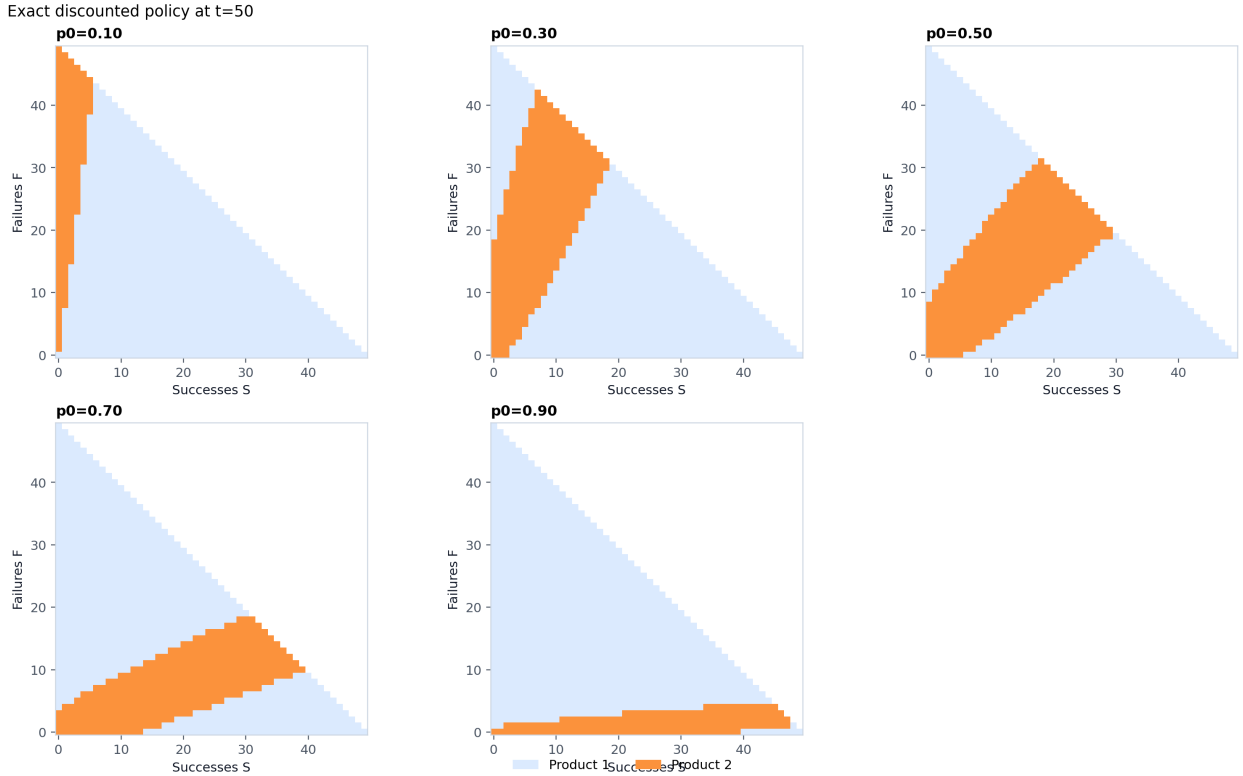


Figure 2: Optimal product choice over the raw count state space (S, F) .

Figure 2 plots the optimal product for Seller A over the state space (S, F) .

- The horizontal axis is the number of observed successes S .
- The vertical axis is the number of observed failures F .
- Blue means product 1, the cheap product.
- Orange means product 2, the high-quality product.
- The blank region is not reachable at the displayed calendar period.

The orange regions are states where a success is sufficiently valuable for future demand. These regions are not simply “bad reputation” or “good reputation” states. Instead, they tend to occur near reputation margins where the next outcome can move the user’s belief enough to change future choice probabilities.

4 Policy Over Posterior State Space

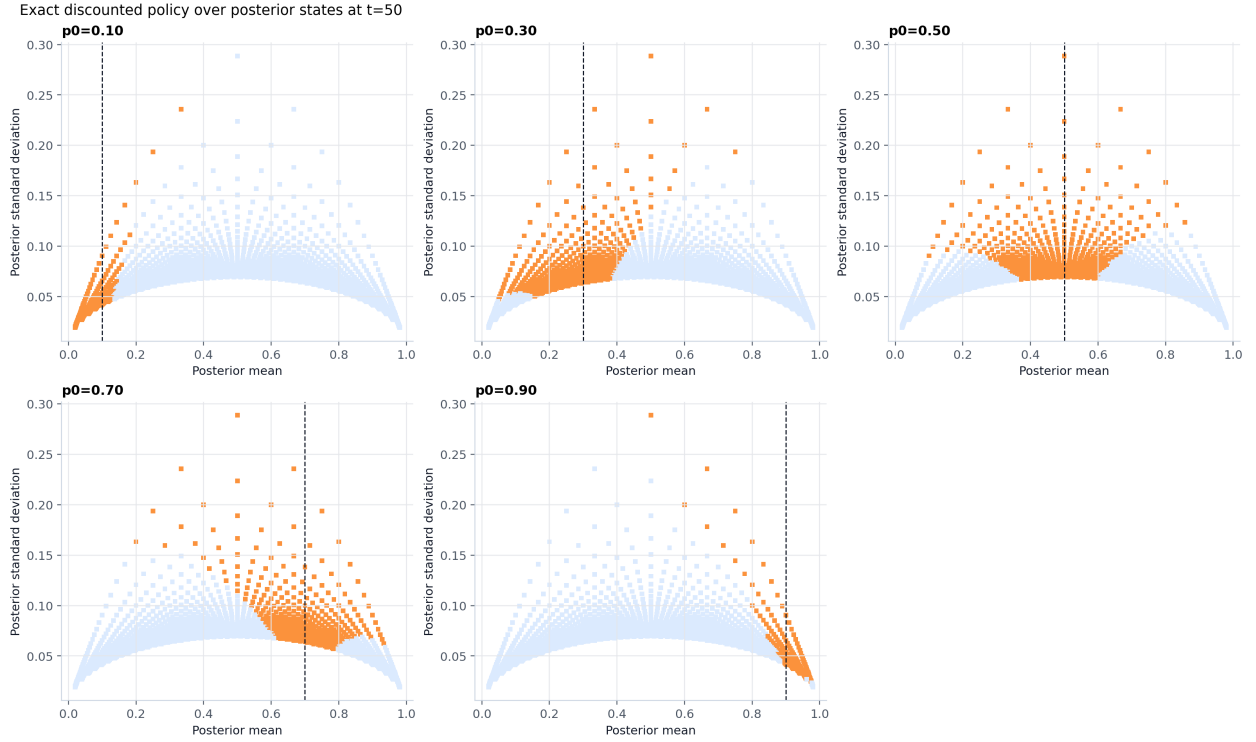


Figure 3: Optimal product choice over posterior mean and posterior uncertainty.

Figure 3 re-expresses the same policy in posterior belief coordinates. Each point is still a count state (S, F) , but it is plotted using

$$m(S, F) = \mathbb{E}[\theta \mid S, F] = \frac{1 + S}{2 + S + F}$$

on the horizontal axis and the posterior standard deviation on the vertical axis.

The dashed vertical line is p_0 . It is the known success probability of Seller B and therefore the Thompson-sampling comparison threshold. The black dot is the initial prior $\text{Beta}(1, 1)$, which has posterior mean 0.5 and standard deviation $\sqrt{1/12}$.

The fan shape is mechanical. As $S+F$ grows, the posterior becomes more precise, so points move downward. At the same time, different success rates generate different posterior means, spreading points horizontally. The upper part of the fan corresponds to low-count, high-uncertainty states; the lower part corresponds to high-count, precise states.

5 Value-Difference Diagnostic

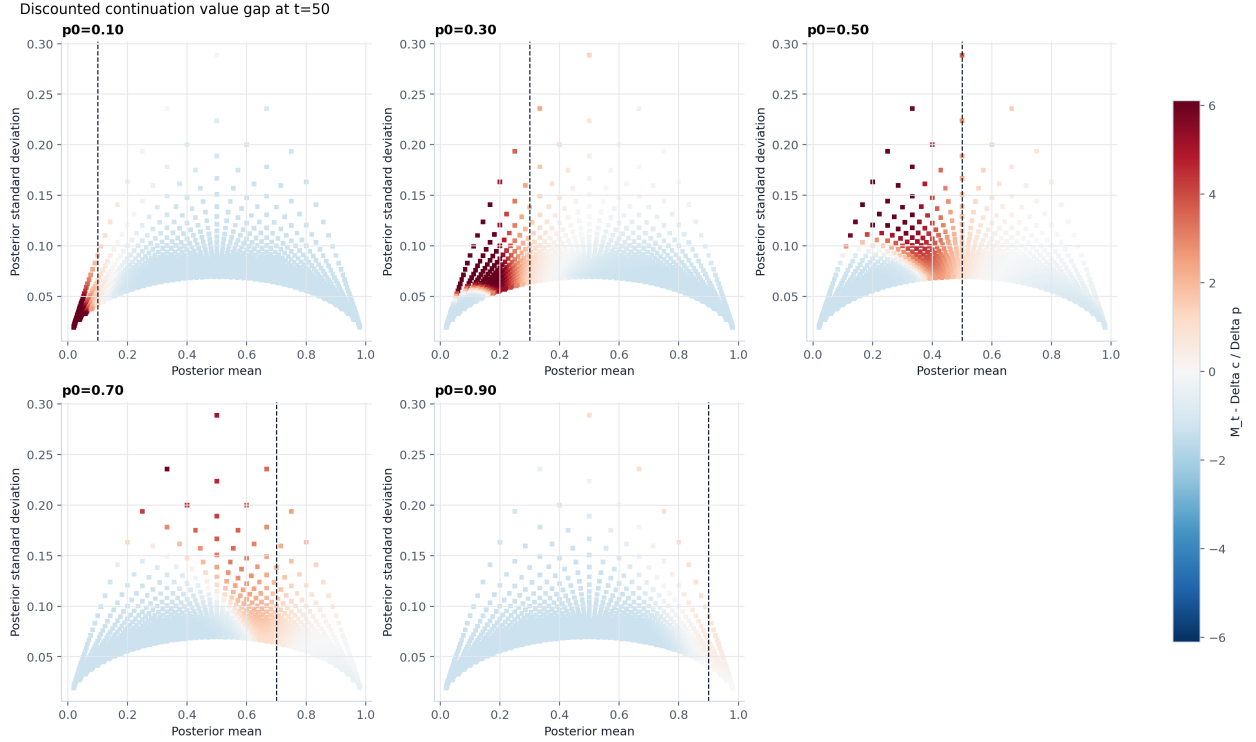


Figure 4: Diagnostic for the discounted continuation-gap threshold $M_t(S, F)$.

Figure 4 plots

$$M_t(S, F) - \frac{c_2 - c_1}{p_2 - p_1}.$$

Positive values mean product 2 is optimal, while negative values mean product 1 is optimal. This plot is useful because it shows the smooth object behind the policy rule. The policy plot discretizes states into blue and orange, while the value-difference plot shows whether those regions come from a coherent threshold or from numerical noise.

The exact backward induction includes every state reachable at the displayed period. It does not impose a fixed observation boundary or project histories back onto a smaller state space.

6 Horizon Convergence

The discounted problem is solved at $T = 700$, where $0.98^{700} = 7.22 \times 10^{-7}$. As a direct check, the code re-solves the problem at $T = 850$ and compares the policies over all 1,275 states reachable at $t = 50$.

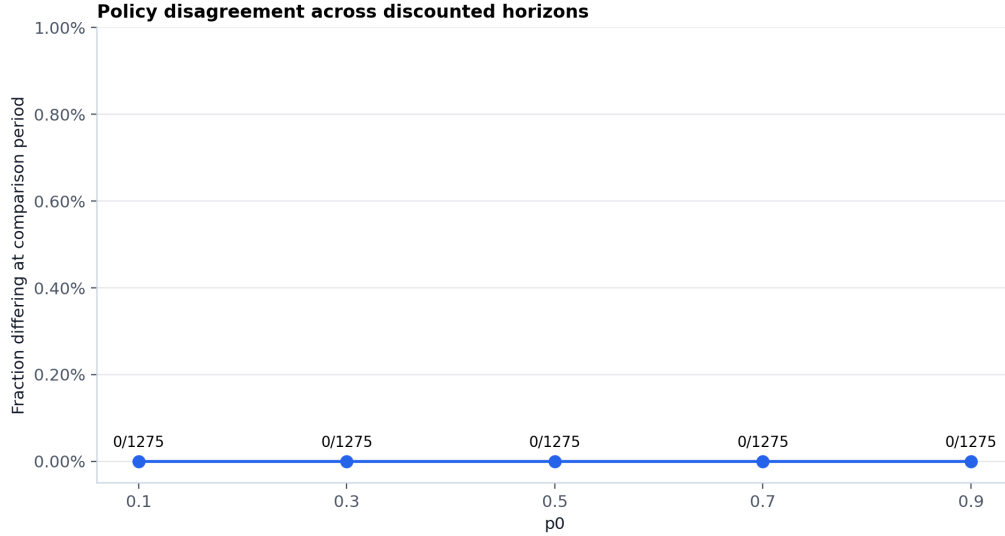


Figure 5: Policy disagreement between $T = 700$ and $T = 850$ at $t = 50$.

Table 1: Policy disagreement between $T = 700$ and $T = 850$ at $t = 50$.

p_0	States compared	Disagreement
0.1	1,275	0%
0.3	1,275	0%
0.5	1,275	0%
0.7	1,275	0%
0.9	1,275	0%

There are no disagreements for any tested p_0 . The terminal horizon is therefore no longer driving the early policy. A direct $p_0 = 0.5$ comparison between $T = 700$ and $T = 2000$ likewise has zero disagreements at $t = 50$.

7 Simulation Path Plot

Figure 6 shows simulated histories under the solved best-response policy. The panels track Seller A's market share, product 2 use conditional on being chosen, per-period profit, and the user's posterior mean.

Figure 7 shows the user's side of the same simulated histories. The posterior mean and posterior standard deviation summarize what the user believes about Seller A before each choice. The demand-probability panel plots $\rho(S, F)$, the probability that the Thompson-sampling user chooses Seller A at that period. The observations panel plots the average number of Seller A outcomes the user has seen.

Exact discounted simulated paths

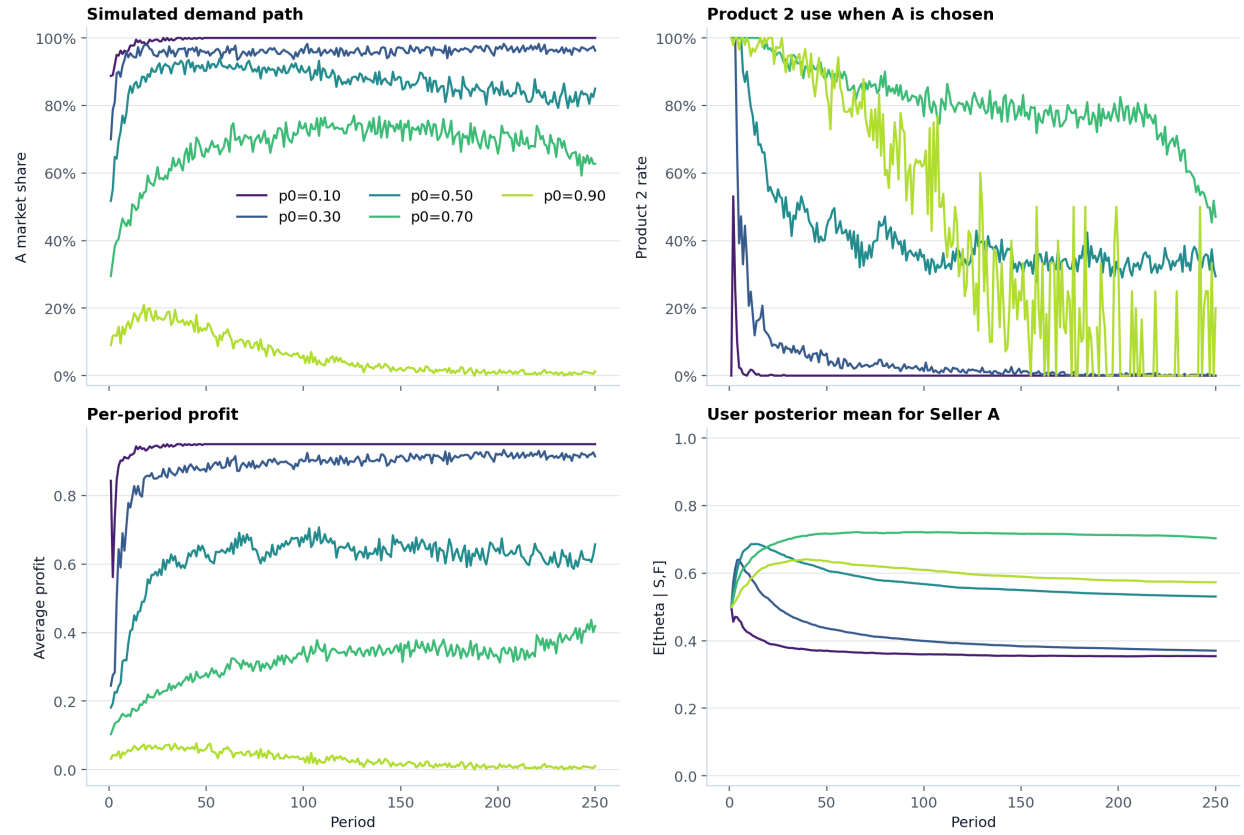


Figure 6: Simulated demand and product 2 usage paths for selected values of p_0 .

Exact discounted user-belief paths

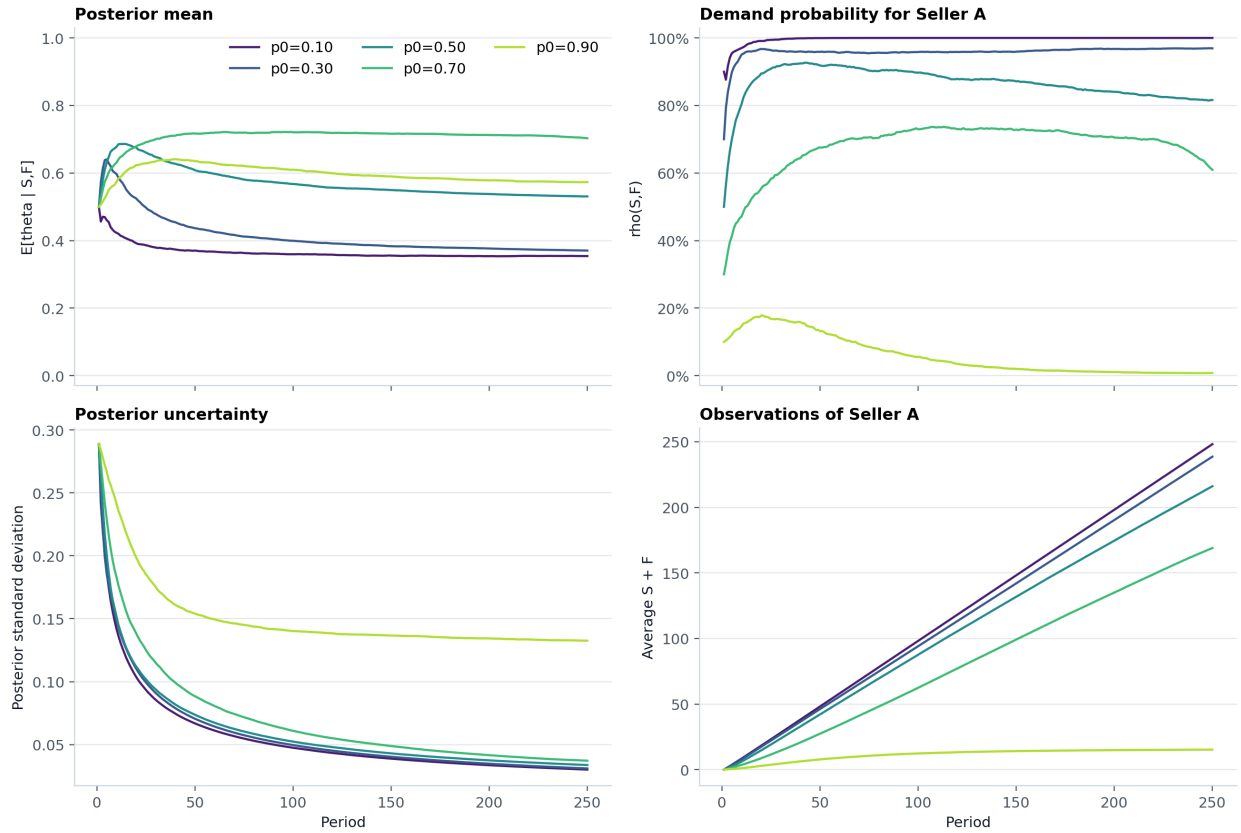


Figure 7: Simulated user belief paths for selected values of p_0 .

As p_0 increases, Seller A is chosen less often. Conditional on being chosen, product 2 use tends to rise over the intermediate range because reputation investment becomes more important. At very high p_0 , demand is so low that investment is often not worthwhile.

A fast decline in product 2 use usually means Seller A has already pushed the user's belief into a safer region. Once the user has many observations of Seller A, posterior uncertainty falls, and one additional success has a smaller effect on $\rho(S, F)$. At that point, the costly product 2 has less continuation value, so the best response can switch back toward product 1 even if demand remains high.